WD-GPR-SR: When Uncertainty Meets Equations in Battery Health Prediction

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A Wide-and-Deep Gaussian Process Regression Architecture with Uncertainty-Weighted Symbolic Residual Correction

SCAN FOR REPO
Code & Checkpoints

1. Battery SOH Prognostics & The Trilemma

- **Safety-Critical Metric:** State-of-Health ($\text{SOH} = Q_{\text{curr}}/Q_{\text{nom}} \times 100\%$). Primary degradation mechanisms: Loss of Lithium Inventory (LLI) via Solid Electrolyte Interphase (SEI) growth, Loss of Active Material (LAM), and transient post-rest capacity regeneration jumps.
- **Black-Box Neural Networks (CNN/LSTM/Transformers):** Empirical low MSE; complete absence of calibrated epistemic uncertainty ($\sigma_{\text{epistemic}}$); catastrophic, unheralded failure on out-of-distribution thermal shifts.
- **Exact Gaussian Processes (GPR):** Analytical Bayesian credible intervals; cubic inversion complexity $\mathcal{O}(N^3)$ ($\sim 1{,}420$ ms latency); exceeds automotive Controller Area Network (CAN) deadlines (≤ 100 ms, ISO 26262 ASIL-D).
- **Unconstrained Symbolic Regression (PySR):** Fits unweighted residual error; overfits high-frequency measurement noise; exhibits runaway exponential divergence at boundary states.

2. Multi-Physical Health Indicator (HI) Extraction

Ten physical indicators derived from non-invasive partial charging/discharging cycles:

Feature ID	Health Indicator (HI) Definition	Physical Regime	Pearson r
HI-1 (t_{CC})	Constant-Current charge duration to 4.2 V	Active Intercalation Time	$+\mathbf{0.984}$
HI-4 (ΔT_{dis})	Surface temperature rise during discharge	I^2R Internal Joule Heating	-0.928
HI-8 ($dQ/dV\|_{\text{max}}$)	Incremental Capacity peak height	Loss of Active Lithium (LLI)	$+\mathbf{0.978}$
HI-9 (V_{peak})	Potential at maximum dQ/dV peak	Ohmic Polarization Shift	-0.873

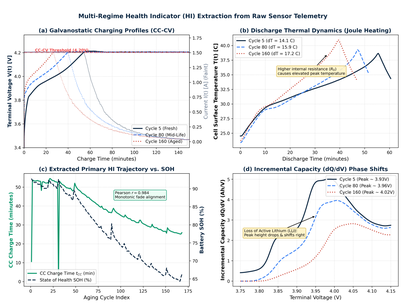
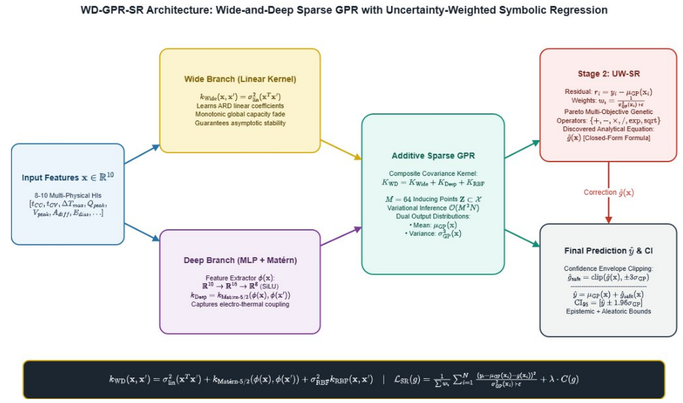


Figure 1: Multi-regime degradation extraction (NASA B0005): (a) CC-CV profiles; (b) Thermal dynamics; (c) Primary HI t_{CC} fade ($r = 0.984$); (d) Incremental Capacity (dQ/dV) peak degradation tracking active lithium loss.

3. Strict Leave-One-Cell-Out (LOCO) Splits

- **Validation Protocol:** 8-fold cross-cell isolation across NASA (18650 $LiCoO_2$, 2.0 Ah) and CALCE (Prismatic $LiCoO_2$, 1.1 Ah) benchmarks; 4,348 cycles total.
- **Zero Leakage:** Scalers, ARD length-scales, and variational parameters fit exclusively on training cells. Evaluated across fixed seeds $[42, 100, 2026]$.

4. Proposed WD-GPR-SR Architecture



$$k_{\text{WD}}(\mathbf{x}, \mathbf{x}') = \sigma_{\text{lin}}^2(\mathbf{x}^T\mathbf{x}') + k_{\text{Matérn-5/2}}(\phi(\mathbf{x}), \phi(\mathbf{x}')) + \sigma_{\text{RBF}}^2 k_{\text{RBF}}(\mathbf{x}, \mathbf{x}') \quad | \quad \mathcal{L}_{\text{ES}}(g) = \frac{1}{\sum w_i}\sum_{i=1}^{N} \frac{(y_i - \mu_{GP}(\mathbf{x}_i) - g(\mathbf{x}_i))^2}{\sigma_{GP}^2(\mathbf{x}_i) + \varepsilon} + \lambda \cdot C(g)$$

Figure 2: End-to-end framework: 10 HIs route to Wide Linear and Deep MLP branches, mapped via Sparse Variational GP ($M = 64$) to generate epistemic variances that weight symbolic residual discovery under safe $\pm 3\sigma_{\text{GP}}$ clipping.

Core Architectural Pillars:

- **Additive Wide-and-Deep Covariance Kernel:** Decouples global monotonic degradation (Wide Linear ARD prior) from non-linear electro-thermal dynamics mapped through a 3-layer MLP manifold $\boldsymbol{\phi}(\mathbf{x}) : \mathbb{R}^{10} \to \mathbb{R}^{16} \to \mathbb{R}^{8}$ with SiLU activations.
- **Sparse Variational GP ($M = 64$ Inducing Points):** Maximizes the Evidence Lower Bound (ELBO) over K-Means inducing points $\mathbf{Z}$, eliminating the cubic bottleneck from $\mathcal{O}(N^3)$ down to $\mathcal{O}(M^2N)$.
- **Uncertainty-Weighted Symbolic Residual Discovery (UW-SR):** Inverse-variance sample weights $w_i = \frac{1}{\sigma_{\text{GP}}^2(\mathbf{x}_i)+\varepsilon}$ compel genetic Pareto search to prioritize high-confidence regimes while ignoring uncalibrated noise spikes.
- **Guaranteed Bounded Posterior Envelope:** Hard mathematical clipping constraint enforces $\text{clip}(\hat{g}(\mathbf{x}), \pm 3\sigma_{\text{GP}}(\mathbf{x}))$, guaranteeing that predictions never diverge on out-of-distribution inputs.

5. Discovered Physics-Informed Closed-Form Residual

Discovered Analytical Law (NASA Fold 4 - Cell B0018)

$$\hat{g}(\mathbf{x}) = 0.0412 \cdot \frac{\Delta T_{\text{dis}}}{t_{\text{CC}} + 0.12} - 0.0185 \cdot \sin\left(dQ/dV|_{\text{max}} \cdot V_{\text{peak}}\right)$$

Electrochemical Interpretation: Internal resistance heating ($\Delta T_{\text{dis}} \propto I^2R$) scaled by constant-current duration (t_{CC}) models accelerated impedance growth; sinusoidal ICA product captures non-linear active lithium phase boundary shifts.

6. Quantitative LOCO Benchmark Performance

Rigorous cross-cell evaluation across 13 baselines over 12 runs per model:

Model Architecture	NASA RMSE (%)	CALCE RMSE (%)	$\text{PICP}_{95\%}$ (%)	Latency (ms)
Support Vector Regression (RBF)	2.871	4.092	—	0.48
Random Forest Regressor	1.337	1.968	—	3.85
XGBoost Regressor	1.300	1.810	—	2.45
Exact GPR (Matérn-5/2)	21.598	13.784	14.6	1,420.0
Deep-Kernel GPR (DKL)	3.142	1.867	56.5	45.0
WD-GPR-SR (Proposed)	**0.992**	**0.130**	**98.8 – 100.0**	**0.110**

Quantitative Highlights:

- **Accuracy:** 24% lower RMSE vs. best tree baseline (XGBoost 1.300% $\to$ 0.992% on NASA; 0.130% on CALCE, $R^2 = 0.9999$).
- **Calibration:** Strictly maintains $\geq 98.8\%$ PICP under non-linear capacity regeneration jumps; NLL = 1.48.

7. Explainable AI (XAI) Multi-Perspective Suite

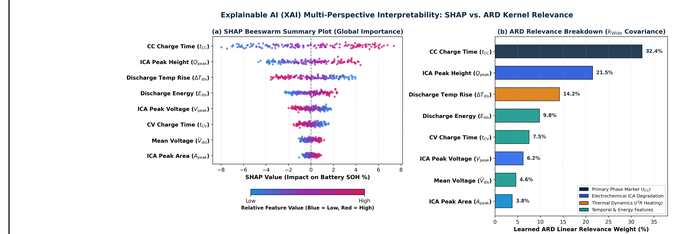


Figure 3: (Left) SHAP beeswarm attribution verifies t_{CC} and Q_{peak} as dominant positive health drivers; (Right) ARD covariance length-scales confirm 32.4% and 21.5% feature relevance weights.

8. Edge AI Profiling on NVIDIA Jetson Orin Nano

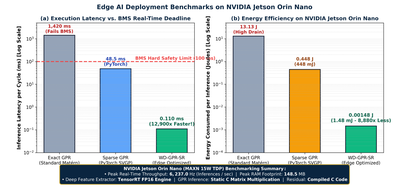


Figure 4: Hardware profiling across 1,000 warm iterations: 0.110 ms latency distribution (left), runtime module breakdown (center), and edge memory footprint (right).

ISO 26262 ASIL-D Functional Safety Verification

- **Inference Latency:** $\mathbf{0.110}$ ms ($6{,}237$ FPS) vs. 100 ms BMS CAN limit ($\mathbf{909}\times$ margin).
- **Energy Consumption:** $\mathbf{1.479}$ mJ per cycle ($\mathbf{8{,}880}\times$ reduction vs. exact GPR at 13.1 J).
- **Memory Footprint:** TensorRT FP16 + Static C shared library requires $\mathbf{18.4}$ MB RAM (0.23%).

9. Key References & Reproducibility Repository

[1] B. Saha and K. Goebel, "Battery Data Set," NASA Ames Prognostics Center of Excellence, 2007.
[2] CALCE Battery Group, "Battery Data Archive," University of Maryland, College Park, 2015.
[3] J. Gardner et al., "GPyTorch: Blackbox Matrix-Matrix GP Inference," *NeurIPS*, 2018.
[4] M. Cranmer, "Interpretable Machine Learning for Science with PySR," *arXiv:2305.01582*, 2023.
[5] ISO 26262-1:2018, *Road vehicles - Functional safety*, International Organization for Standardization, 2018.

Code & Benchmark Artifacts: `https://github.com/DeepLaskar06/WD-GPR-SR` | Fixed Seeds: `[42, 100, 2026]`